Part 2: Evaluation Strategy for a Text-to-SQL Agent

Objective

This document proposes a structured evaluation framework for a Text-to-SQL AI agent that understands natural language, generates SQL queries, executes them, and returns answers. The goal is to systematically measure whether the agent is actually performing well not just generating SQL, but returning correct, reliable, and timely results.

1. Evaluation Data

- Use the benchmark questions from Part 1 with verified ground-truth SQL queries.
- Keep the same database schema and seed data (seed.sql) to ensure consistent and reproducible results.
- Optionally include a small set of paraphrased questions (same meaning, different wording) and ambiguous questions to test robustness.

2. Core Metrics

A. Execution Accuracy (Primary Metric)

Definition: The SQL generated by the agent returns the same result as the ground-truth SQL.

Method: Execute both the agent-generated SQL and the ground-truth SQL. Compare the result sets for an exact match (same rows, same columns, same values).

B. SQL Validity

Definition: The generated SQL runs without any syntax or runtime errors.

Method: Track the error rate across all test questions. Log cases involving syntax errors, missing columns, or missing tables.

C. Component Accuracy

This checks whether individual parts of the generated SQL are correct:

- **Table Selection Accuracy** - Are the correct tables referenced?
- **Column Selection Accuracy** - Are the correct columns selected or filtered?
- **Join Correctness** - Are the correct join keys used? Are all required joins present?

Method: Parse the generated SQL and compare its components (tables, columns, joins) against the ground-truth SQL components.

D. Result Quality

Definition: The answer returned to the user is correct and complete.

Method: Compare result sets between agent output and ground truth. For aggregation queries (COUNT, SUM, AVG, etc.), compare numeric outputs directly.

E. Latency

Definition: Total time from receiving the natural language question to returning the final answer.

Method: Log total response time and SQL execution time separately. Calculate the average latency and the p95 latency (the time within which 95% of queries are answered).

3. Robustness Tests

To measure how well the agent handles real-world variation:

- **Paraphrase Questions:** Same question asked in different words. The agent should return the same result.
- **Ambiguous Questions**: Questions where multiple interpretations are valid. Evaluate whether the agent picks a reasonable interpretation or asks for clarification.
- **Missing Information Prompts**: Questions intentionally missing key details. Check if the agent asks a clarifying question instead of guessing.

4. Error Handling and Self-Correction

- Measure the retry success rate: when the agent fails on the first attempt, does it successfully correct itself on retry?
- Log the number of retries per question.
- Record whether the final result (after retries) matches the ground truth.

5. Natural Language Answer Quality

Beyond just SQL correctness, evaluate the final answer presented to the user:

- Is the response readable and clearly written?
- Does the answer directly address what was asked (e.g., does it explain whether the result is a list, a count, or a grouped summary)?
- Is the response free of misleading or irrelevant information?

6. Evaluation Procedure

For each benchmark question, follow these steps:

1. Submit the natural language question to the agent.
2. Capture the SQL generated by the agent.
3. Execute the agent's SQL and record the output.
4. Execute the ground-truth SQL and record the output.
5. Compare both outputs to determine execution accuracy (pass/fail).
6. Log any SQL errors, component mismatches (wrong tables, columns, joins), and latency.

After all questions are evaluated, summarize results across all metrics.

7. Acceptance Criteria

The agent is considered production-ready when it meets all of the following thresholds:

Metric	Minimum Threshold
Execution Accuracy	≥ 80%
SQL Validity	≥ 90%
Retry Success Rate	≥ 50% on failed first attempts
Average Latency (simple queries)	< 2 seconds

8 Tools and Artifacts

To ensure reproducibility and traceability, store the following for every evaluation run:

- All natural language questions used for testing
- All agent-generated SQL queries
- All query execution outputs (agent and ground truth)
- All error logs (syntax errors, runtime failures, retries)
- Ground-truth SQL and expected outputs as the reference baseline

Conclusion

This evaluation framework provides a comprehensive and repeatable way to measure the quality of a Text-to-SQL agent across six dimensions: correctness, validity, component accuracy, result quality, robustness, and latency. It can be reused in later tasks to compare different model versions, prompt strategies, or system configurations, making it a reliable foundation for continuous improvement.